INFO8006: Project 2 - Report

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1 Bayes filter

a. The sensor measurement $E_t \in \mathbb{R}$ represents a noisy Manhattan distance between Pacman at position P_t and a ghost at position X_t. The measurement equation is defined as:

$$E_t = d(X_t, P_t) + W_t \tag{1}$$

where $d(X_t, P_t) = \|X_t - P_t\|_1$ denotes the exact Manhattan distance, and W_t is the zero-mean additive noise generated from a shifted Binomial random variable $K \sim \text{Binomial}(n, p)$:

$$W_t = K - np \tag{2}$$

with parameters $p = 0.5$ and $n = \lfloor \frac{\sigma^2}{p(1-p)} \rfloor$, where σ^2 is the sensor variance.

Given that $K = E_t - d(X_t, P_t) + np$, the sensor probability model $P(E_t = e \mid X_t = x)$ evaluates to the probability mass function of the binomial distribution evaluated at integer $k = \text{round}(e - d(x, P_t) + np)$:

$$P(E_t = e \mid X_t = x) = \begin{cases} \binom{n}{k} p^k (1-p)^{n-k}, & \text{if } 0 \le k \le n \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

b. A unified, parameterized transition model for all three ghost variants (`scared`, `afraid`, and `confused`) can be established using a single non-negative parameter $\gamma \ge 0$.

Let $\mathcal{N}(x)$ denote the set of valid adjacent grid positions accessible from current position x via legal cardinal directions (North, South, East, West, excluding maze walls). Each candidate destination $x' \in \mathcal{N}(x)$ is weighted based on its Manhattan distance to Pacman P_t:

$$w(x') = (d(x', P_t))^\gamma \tag{4}$$

The transition probability $P(X_{t+1} = x' \mid X_t = x)$ is given by normalizing these weights:

$$P(X_{t+1} = x' \mid X_t = x) = \frac{(d(x', P_t))^\gamma}{\sum_{y \in \mathcal{N}(x)} (d(y, P_t))^\gamma} \tag{5}$$

If $\mathcal{N}(x) = \emptyset$, the ghost remains at the same position with probability 1. The three ghost behaviors are uniquely derived by setting γ:

- **Confused:** $\gamma = 0$ (Uniform distribution across all legal neighbors).
- **Afraid:** $\gamma = 2.0$ (Quadratic bias towards moves increasing separation).
- **Scared:** $\gamma = 8.0$ (High exponential bias selecting the maximal separation).

2 Implementation

a. ***Leave empty.***

3 Experiment

a. To quantify and summarize the uncertainty of Pacman's belief state B_t over all accessible maze cells $\mathcal{X}$, we use the **Shannon Entropy**:

$$H(B_t) = -\sum_{x \in \mathcal{X}, B_t(x) > 0} B_t(x) \log_2 B_t(x) \tag{6}$$

Entropy attains its maximum when the probability distribution is uniform (highest uncertainty) and approaches zero when the probability mass collapses onto a single known cell.

b. Assuming access to the ground truth ghost position X_t^*, the tracking quality of the belief state is measured via the **Mean Distance Error**:

$$\text{Error}(B_t) = \sum_{x \in \mathcal{X}} B_t(x) \|x - X_t^*\|_1 \tag{7}$$

This metric computes the expected distance between the estimated belief distribution and the ghost's actual position, where lower values signify higher accuracy.

c. The experimental evaluation was conducted on both `large_filter` and `large_filter_walls` across the three ghost behaviors. Figure **??** presents the averaged metrics with error bars representing one standard deviation.

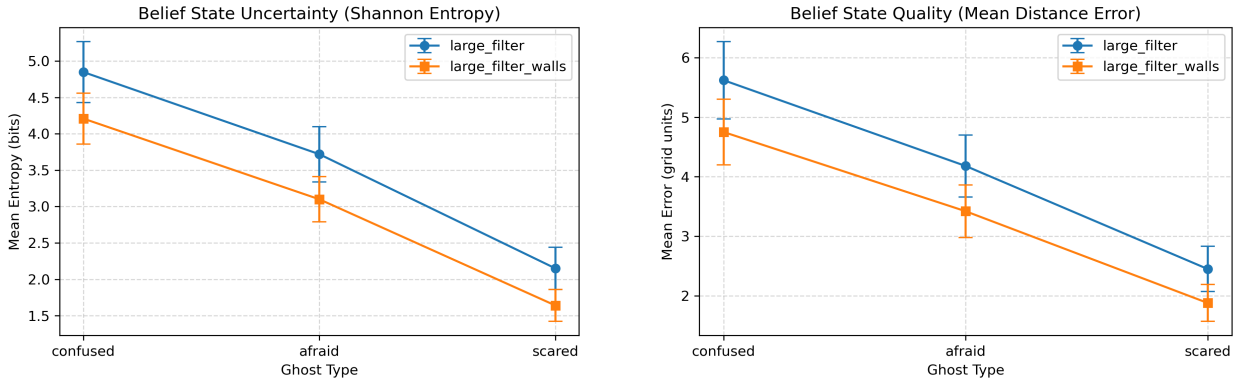


Figure 1: Belief state uncertainty (Entropy) and quality (Mean Distance Error) across ghost types and maze layouts.

d. **Effect of Transition Model Parameter (γ):** As γ increases from 0 (`confused`) to 8 (`scared`), the ghost transitions from stochastic exploration to deterministic fleeing. When γ is large, the filter's motion prediction step aligns closely with the ghost's actual trajectory away from Pacman, leading to faster probability concentration and lower steady-state entropy. On layouts with maze walls (`large_filter_walls`), corridor constraints limit the branching factor, which further accelerates belief convergence and suppresses tracking errors compared to open layouts.

e. **Effect of Sensor Variance (σ^2):** A higher sensor variance increases the parameter $n = \lfloor \frac{\sigma^2}{p(1-p)} \rfloor$, thereby broadening the binomial distribution of the noise. Consequently, the likelihood function $P(E_t \mid X_t)$ becomes more dispersed and assigns comparable probabilities to a wider ring of candidate positions around Pacman. This reduces the informative gain of each sensor update, causing the filter to converge more slowly and maintain higher average entropy.

f. A controller for Pacman to capture invisible ghosts can be designed using a **Maximum A Posteriori (MAP)** pursuit strategy. At each step, Pacman identifies the most probable ghost location $\hat{X}_t = \arg\max_x B_t(x)$ from the current belief distribution and evaluates legal actions to greedily minimize the Manhattan distance toward $\hat{X}_t$.

g. ***Leave empty.***